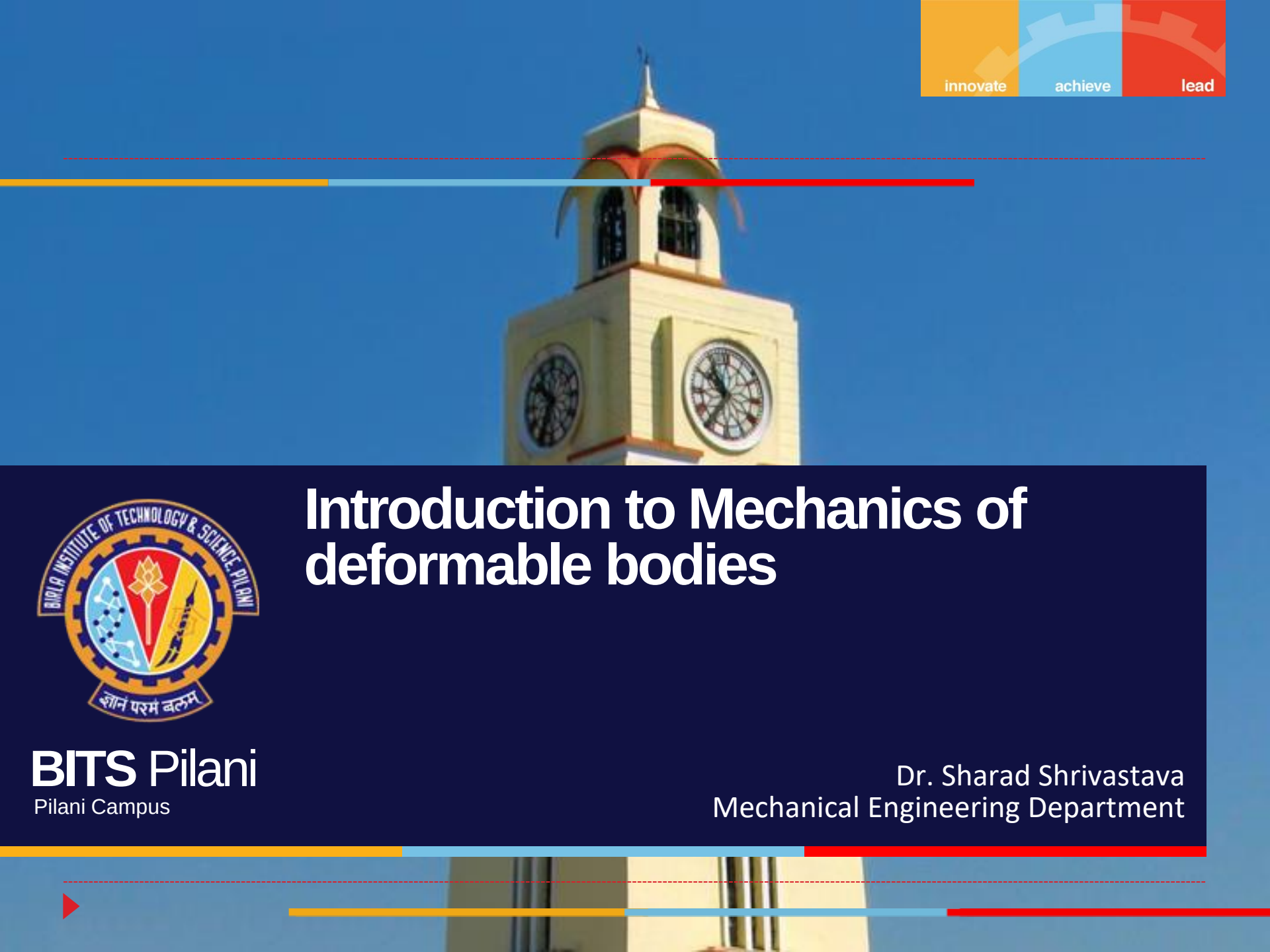
innovate achieve lead



Introduction to Mechanics of deformable bodies

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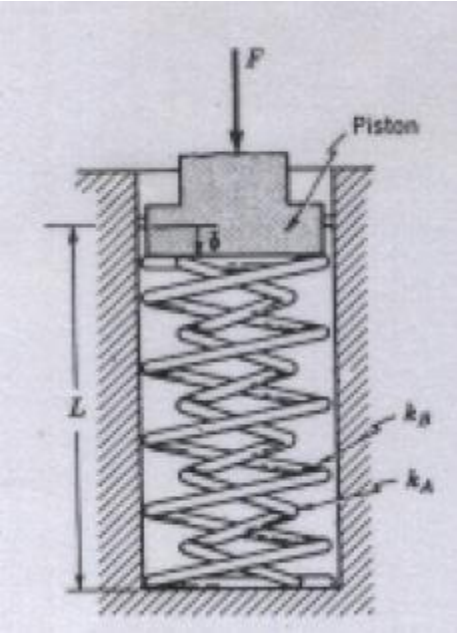
Analysis of deformable bodies

To analyze deformable bodies, 3 steps are required:

- Study of force and equilibrium requirements
- Study of deformation and conditions of geometric fit
- Force – deformation relation

PROBLEM

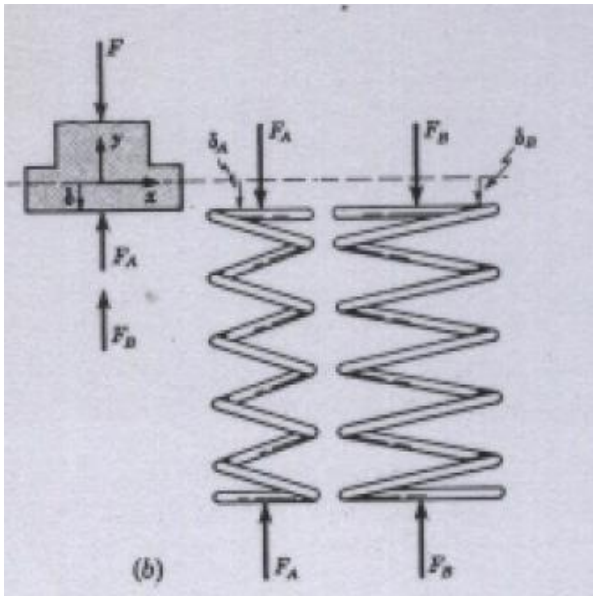
A machine part carrying a load F terminates in a piston which fits in to a cavity as shown in fig. Within the cavity are two springs arranged coaxial with each other. Each spring has the characteristic that the force required to deflect it is proportional to the amount of deflection. Such a spring is said to have a linear force-deflection relation, and the amount of force required to produce a unit deflection is called the spring constant of the spring. We use the symbols K_A and K_B to denote the spring constants of the two springs in the cavity. Calculate how much of the load F is carried by the spring with constant K_A.



Model :
Piston and two springs

Assumption.:
The ends of spring are flat, such that resultant compressive force on the springs are along its axis

STEP 1: STUDY OF FORCES AND EQUILIBRIUM REQUIREMENTS



$$\sum F_Y = 0 : \quad F = F_A + F_B \text{ for piston} \quad \text{---------- (1)}$$

$$\sum F_Y = 0 : \quad \text{already satisfied for Springs}$$

$$\sum M = 0 : \text{already satisfied for both springs and pistons}$$

STEP 2: STUDY OF DEFORMATION AND GEOMETRIC COMPATIBILTY (GEOMETRIC FIT)

Here the action of piston is to cause both the springs to move by the same amount as piston

Condition of geometrical compatibility

$$\delta_A = \delta_B = \delta \text{ ----------(2)}$$

STEP 3: FORCE- DEFORMATION RELATION

$$Given : F \propto \delta$$

$$F_A = K_A \delta_A$$
$$F_B = K_B \delta_B \quad \text{----------(3)}$$

Substitute in eq1

$$F = F_A + F_B$$
$$F = K_A \delta_A + K_B \delta_B$$

Also from eq 2 $\boldsymbol{\delta_A = \delta_B = \delta}$

$$F = (K_A + K_B)\delta_A \text{----------(4)}$$

Dividing eq 4 by first of eq 3 i.e $F_A = K_A \delta_A$

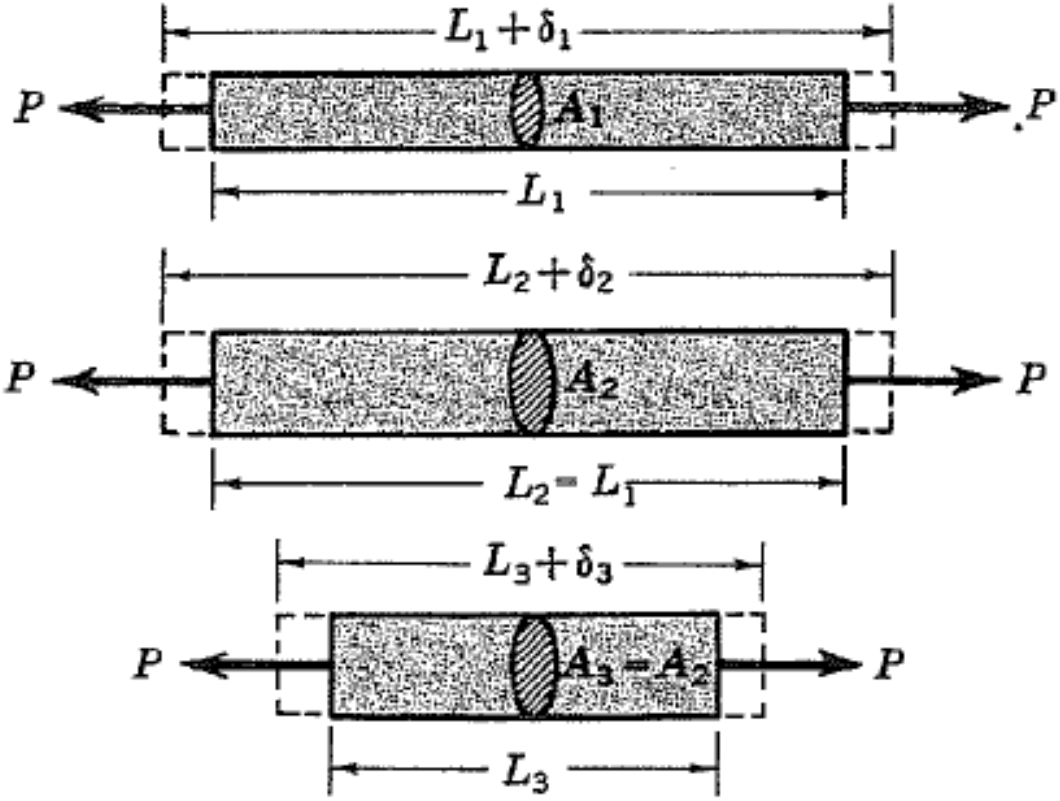
$$\frac{F}{F_A} = \frac{(K_A + K_B)\delta_A}{K_A \delta_A}$$

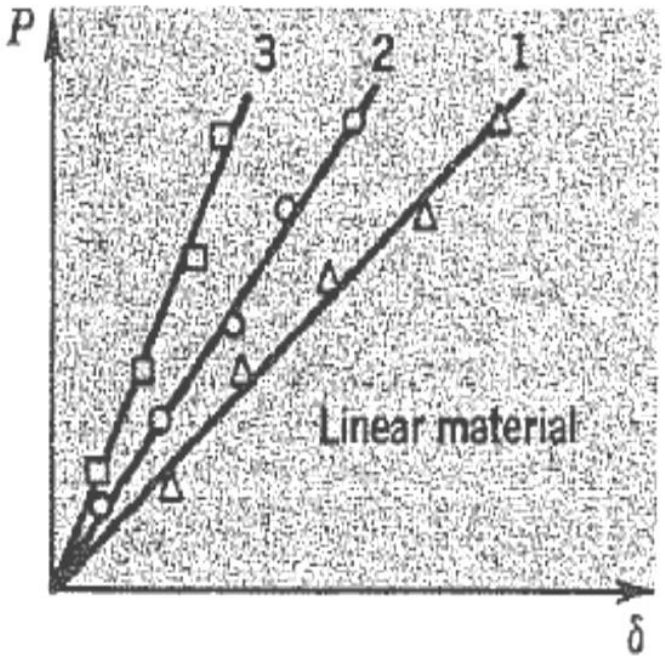
$$\frac{F}{F_A} = \frac{(K_A + K_B)}{K_A}$$

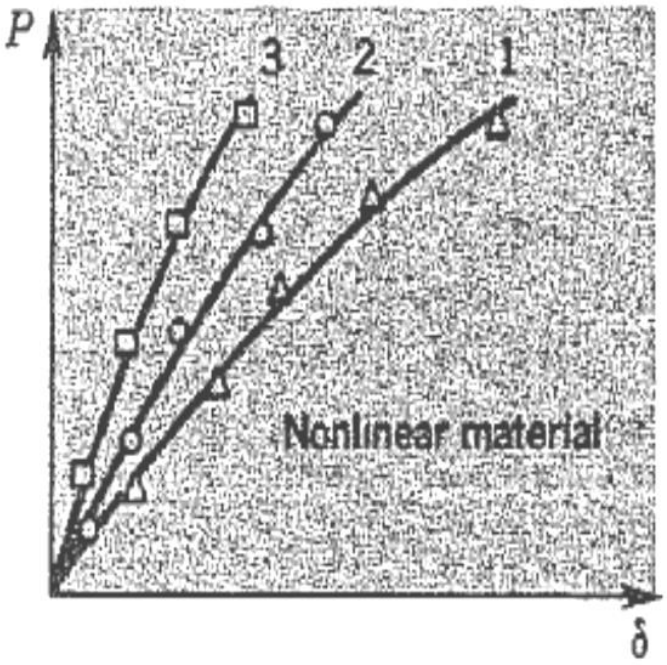
Total deflection of piston

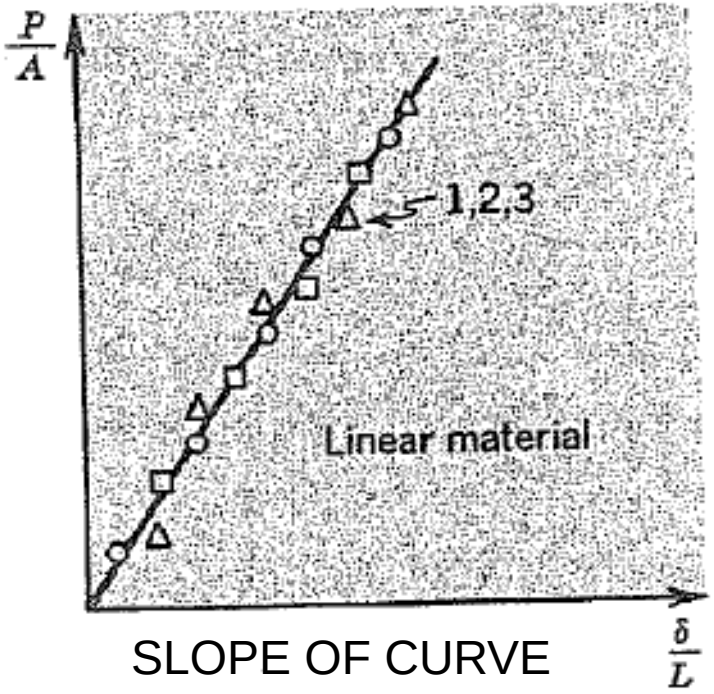
$$\delta = \frac{F_A}{K_A} = \frac{F}{(K_A + K_B)}$$

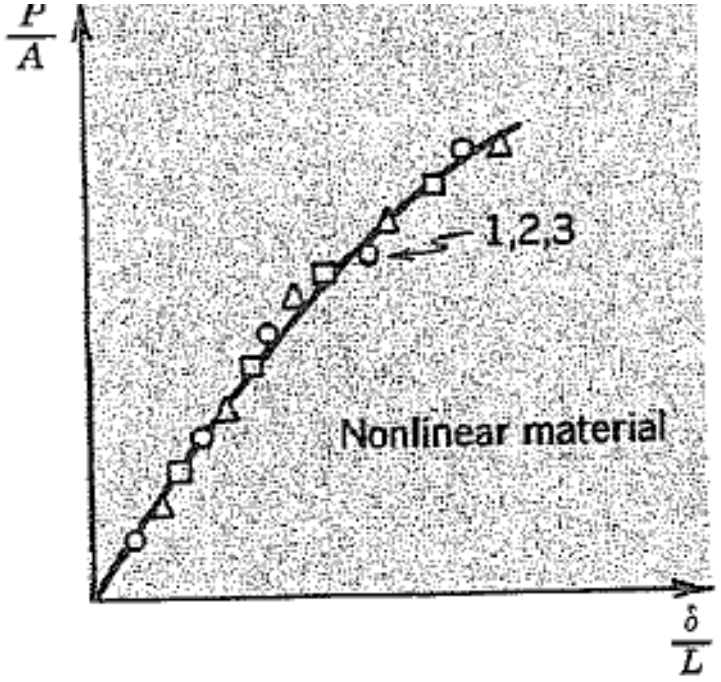
Uniaxial Loading and Deformation




P
3
2
1
Linear material
δ


P
3
2
1
Nonlinear material
δ





SLOPE OF CURVE

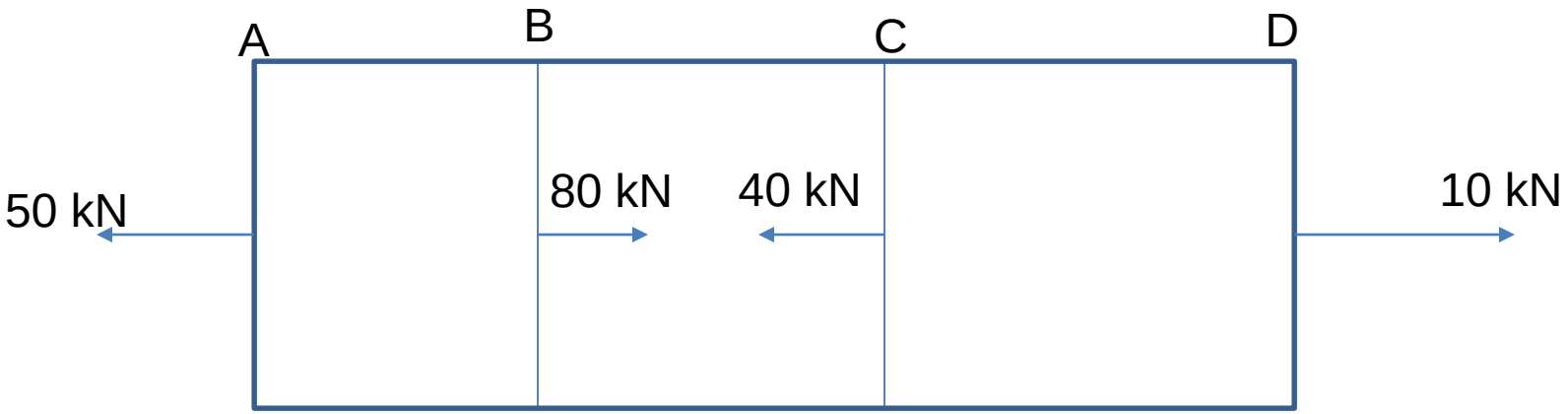
$$E = \frac{PL}{A\delta}$$

$$\delta = \frac{PL}{AE}$$

Modulus of elasticity

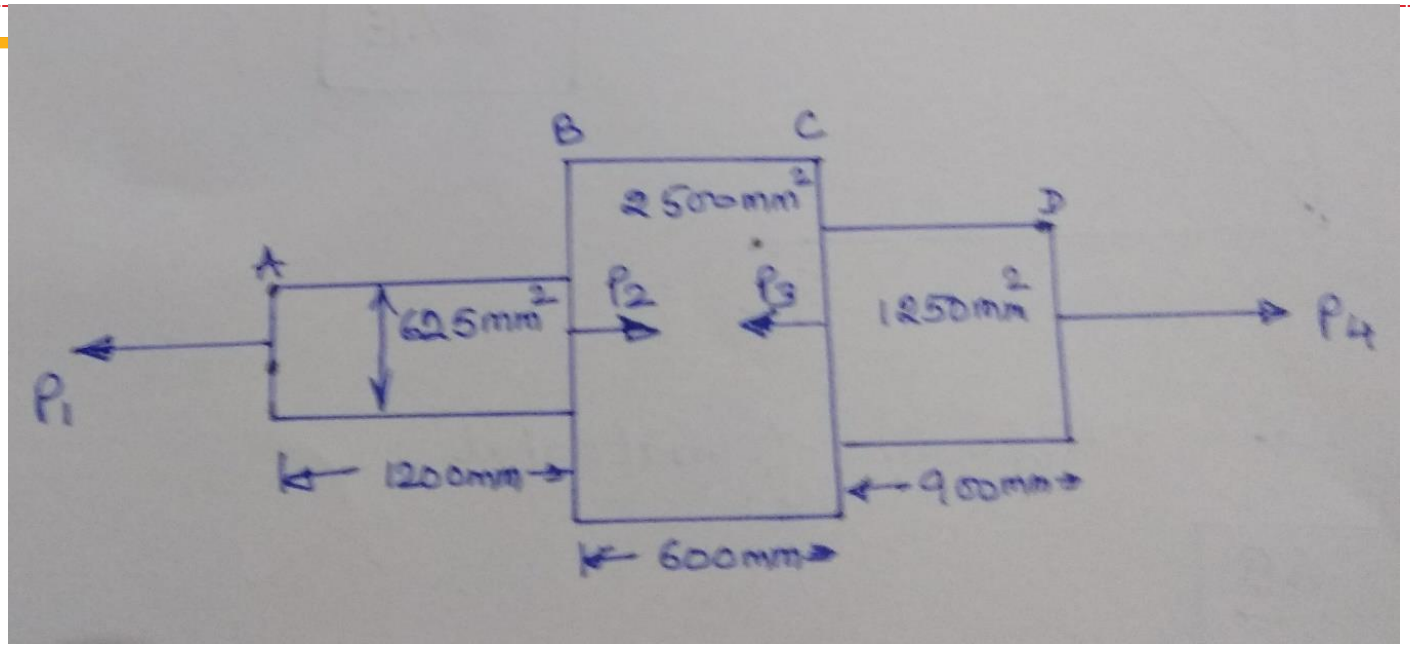
E → represents Stiffness of material

Changes in length under non-uniform conditions



A brass bar having cross-sectional area of 1000mm^2 is subjected to axial forces as shown in fig. Find the total elongation of the bar.
Given: E= 1.05X10^5 N/mm^2 , L_{AB}= 600mm, L_{BC}= 1m, L_{CD}= 1.2m,

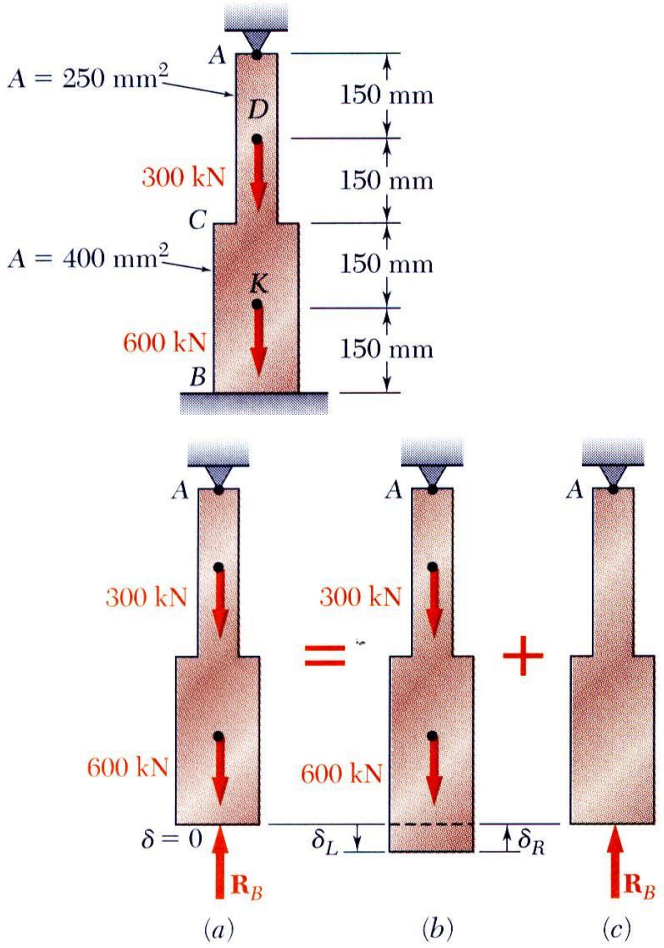
Answer: 0.2857mm



A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in fig. Calculate the force P_2 necessary for equilibrium, if P_1=45kN, P_3=450kN and P_4=130kN. Determine the total elongation of the member, assuming the modulus of elasticity to be $2.1X10^5$ N/mm^2

$P_{2=}$ 365 kN , 0.4903 m(extension)

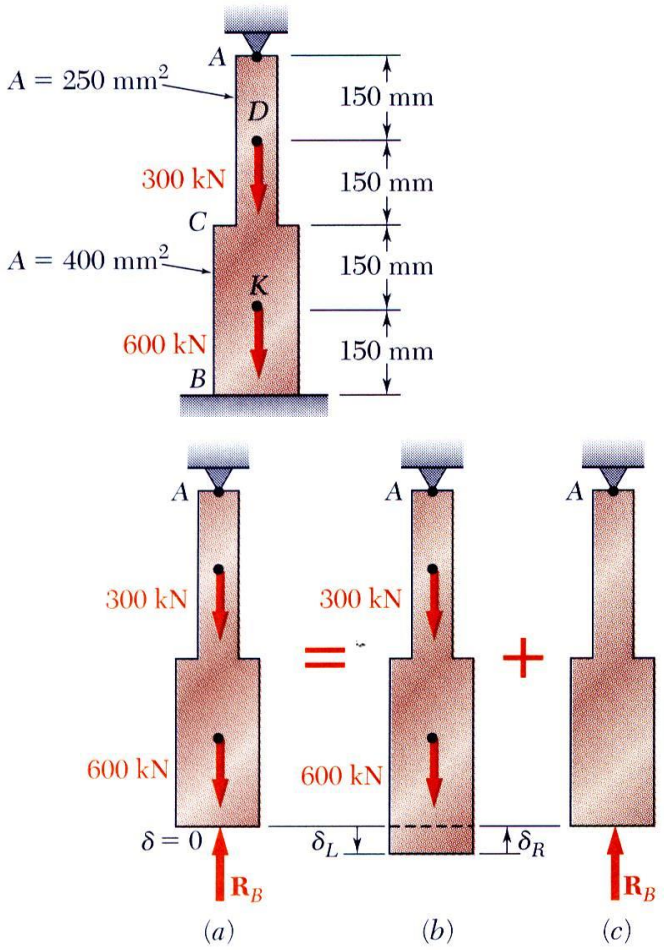
Static Indeterminacy



- Structures for which internal forces and reactions cannot be determined from statics alone are said to be *statically indeterminate*.
- Redundant reactions (supports) are replaced with unknown loads which along with the other loads must produce compatible deformations.
  - Deformations due to actual loads and redundant reactions are determined separately and then added or *superposed* (method of superposition)

$$\delta = \delta_L + \delta_R = 0$$

Static Indeterminacy

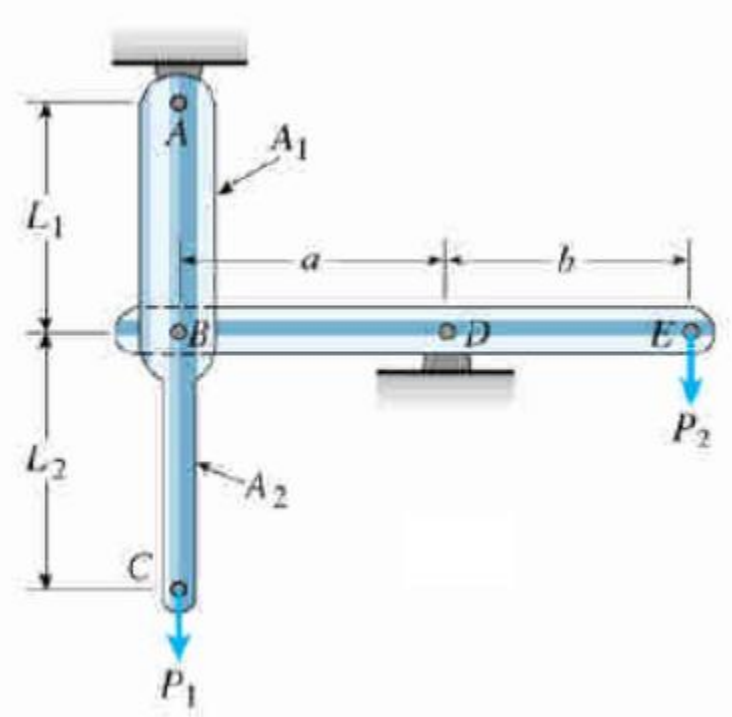


Determine the reactions at *A* and *B* for the steel bar and loading shown, assuming a close fit at both supports before the loads are applied.

SOLUTION:

- Consider the reaction at *B* as redundant, release the bar from that support, and solve for the displacement at *B* due to the applied loads.
- Solve for the displacement at *B* due to the redundant reaction at *B.*
- Require that the displacements due to the loads and due to the redundant reaction be compatible, i.e., require that their sum be zero.
- Solve for the reaction at *A* due to applied loads and the reaction found at *B*.

Ex:



A vertical steel bar ABC has lengths and areas as given in fig. A load P_1 acts at point C. A horizontal arm BE is pinned to the vertical bar at B and carries a load P_2 at E. Calculate the vertical deflection δ at point C if,
P_1 =10kN, P_2 =26 kN, a=b, L_1 =0.5m, A_1 =160mm^2 , L_2 =0.8m, A_2 =100mm^2 , E= 200GPa for steel.

Neglect the weight of the bar.

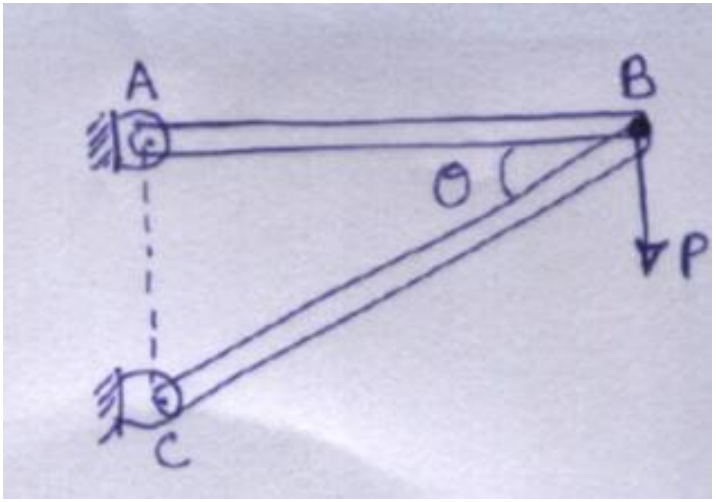
Displacement diagram

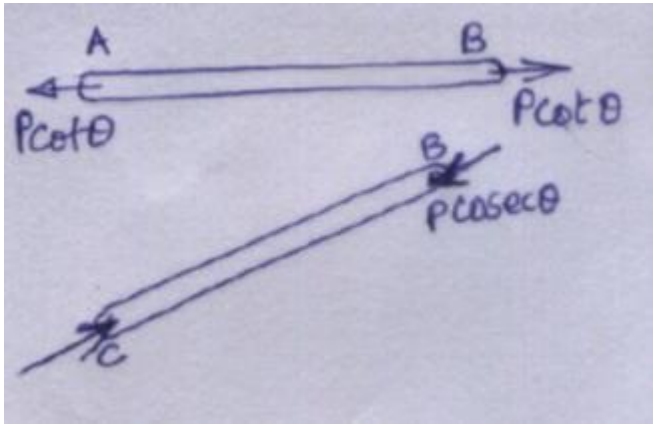
So far we have discussed the procedure for finding the changes in length of single axially loaded members

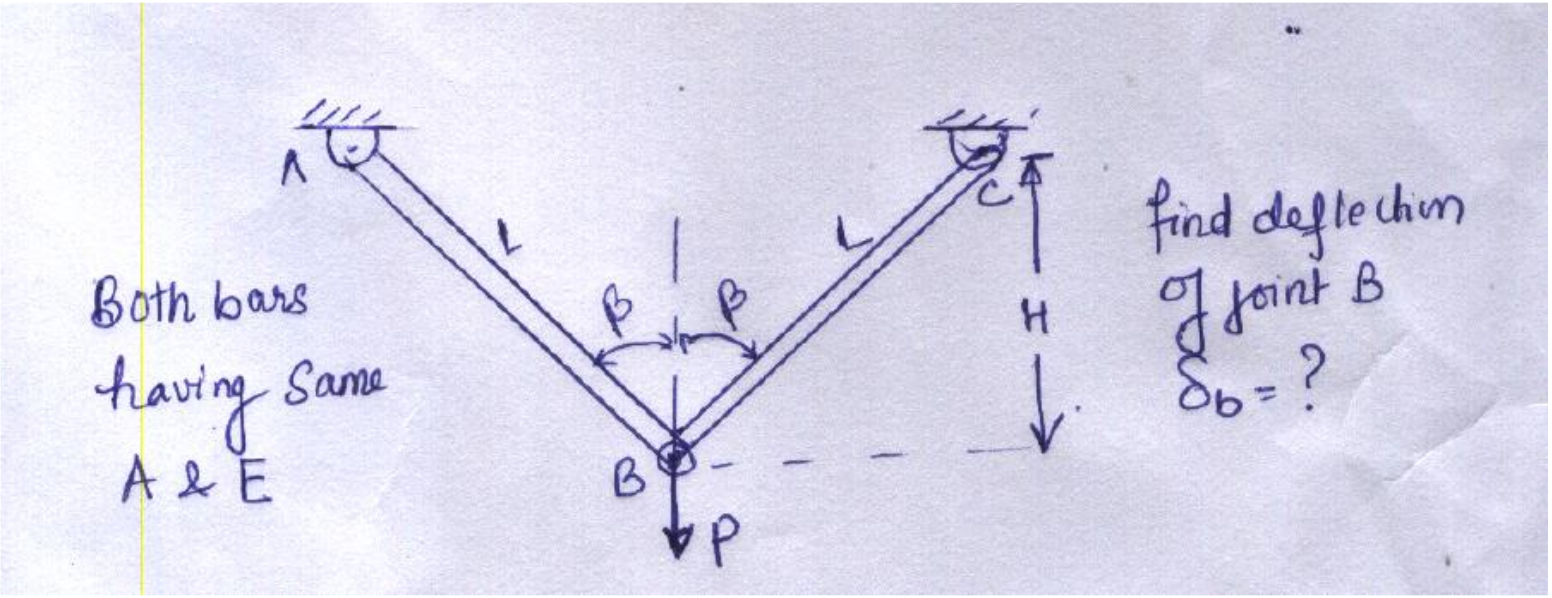
When the structure contains 2 or more than 2 axially loaded members having pinned connections at their ends then we can use displacement diagrams for determining displacements.



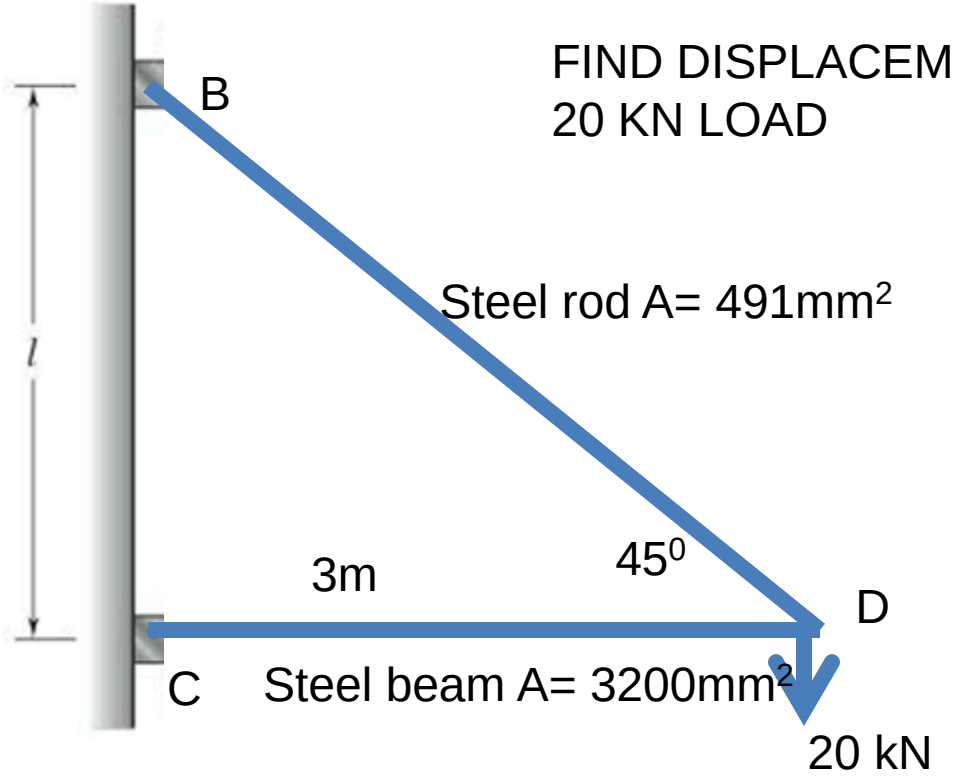
FIND THE DEFLECTION OF JOINT "B" DUE TO VERTICAL LOAD P





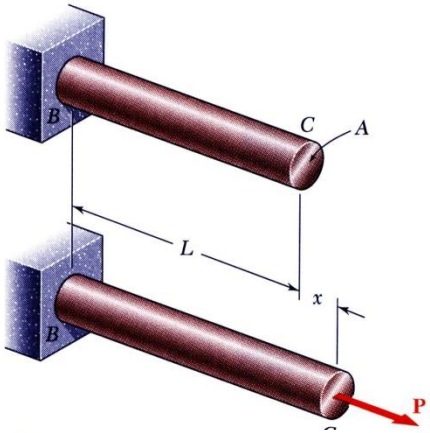

A
L
L
C
H
B
β
β
B
P
Both bars
having Same
A & E
find deflection
of joint B
δ_b = ?

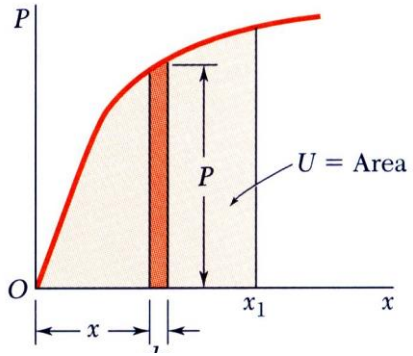
FIND DISPLACEMENT OF POINT D DUE TO 20 KN LOAD

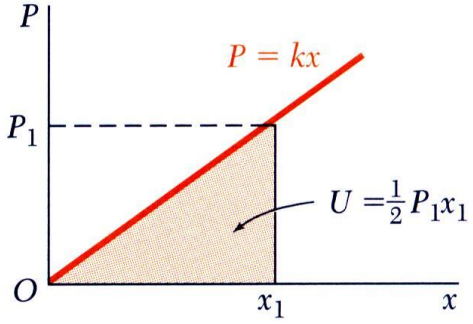


Given l= 3m, E=205GPa

Strain Energy







- A uniform rod is subjected to a slowly increasing load
- The *elementary work* done by the load P as the rod elongates by a small *dx* is

$$dU = P\,dx = elementary\ work$$

which is equal to the area of width *dx* under the load-deformation diagram.

- The *total work* done by the load for a deformation x_1,

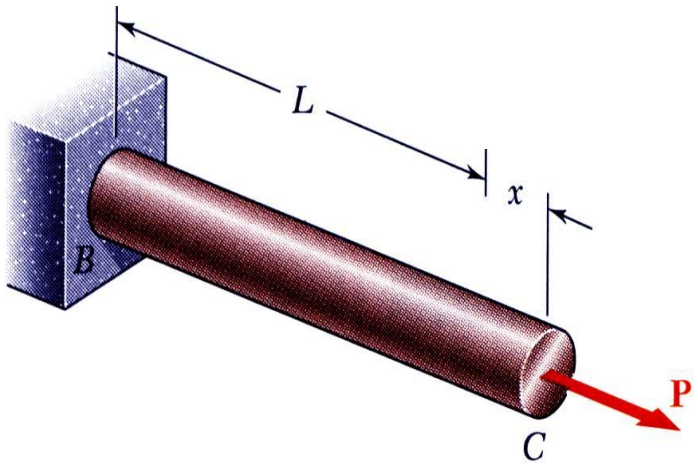
$$U = \int_0^{x_1} P\,dx = total\ work = strain\ energy$$

which results in an increase of *strain energy* in the rod.

- In the case of a linear elastic deformation,

$$U = \int_0^{x_1} kx\,dx = \frac{1}{2}kx_1^2 = \frac{1}{2}P_1x_1$$

Work and Energy Under a Single Load



- Knowing the relationship between force and displacement,

$$x_1 = \frac{P_1 L}{AE}$$

$$U = \frac{1}{2} P_1 \left(\frac{P_1 L}{AE} \right) = \frac{P_1^2 L}{2AE}$$

Castigliano's Theorem

When forces act on elastic systems subject to small displacements, the displacement corresponding to any force, in the direction of the force, is equal to the partial derivative of the total strain energy with respect to that force.

$$\delta_i = \frac{\partial U}{\partial F_i}$$

Where:

F_i = Force at i-th application point

δ_i = Displacement at i-th point in the direction of F_i

U = Total strain energy

Utilizing a Fictitious Force

Castigliano's method can be used to find a deflection at a point even if there is no force applied at that point.

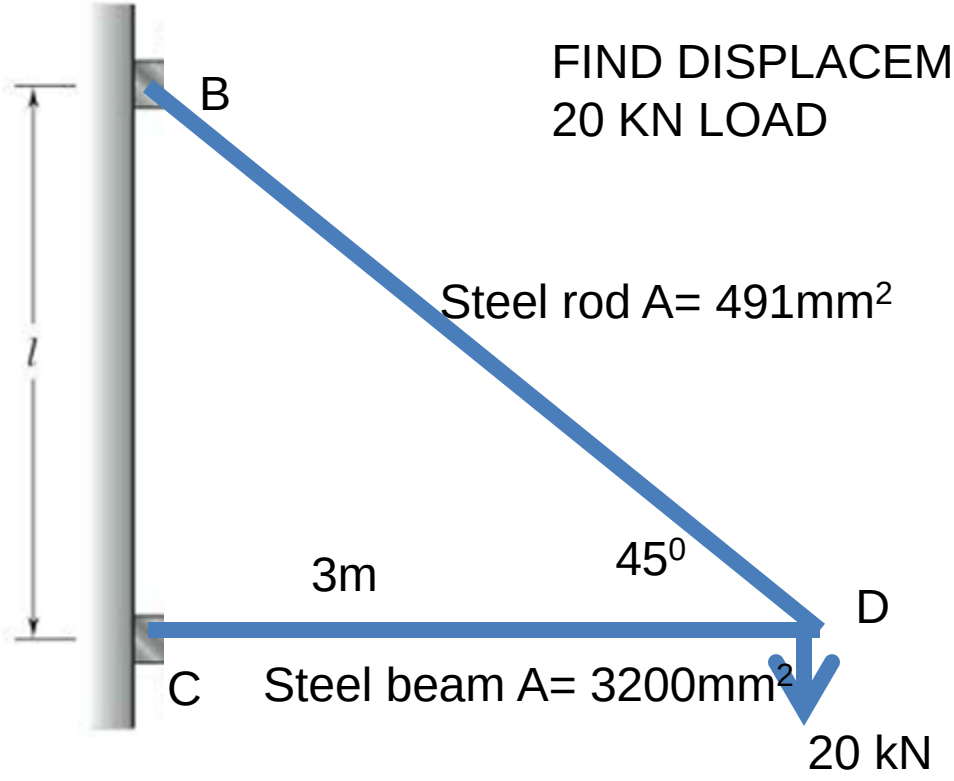
Apply a fictitious force *Q* at the point, and in the direction, of the desired deflection.

Set up the equation for total strain energy including the energy due to *Q*.

Take the derivative of the total strain energy with respect to *Q*.

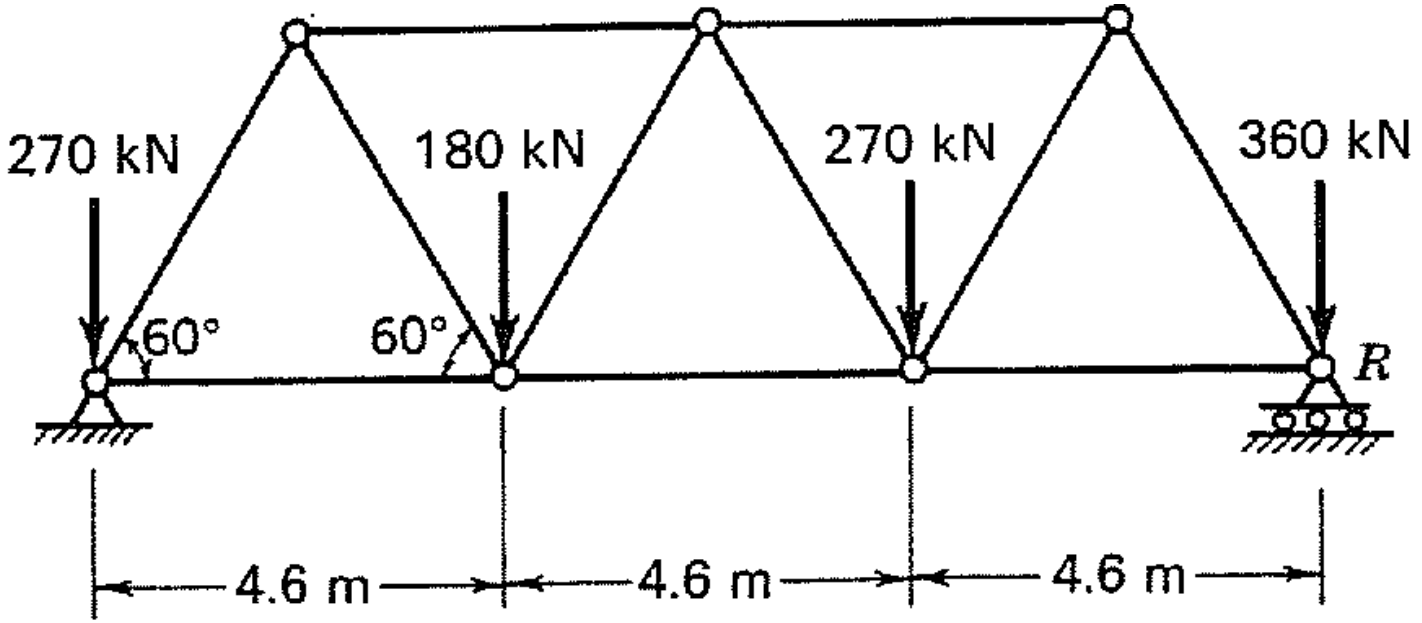
Once the derivative is taken, *Q* is no longer needed and can be set to zero.

FIND DISPLACEMENT OF POINT D DUE TO 20 KN LOAD

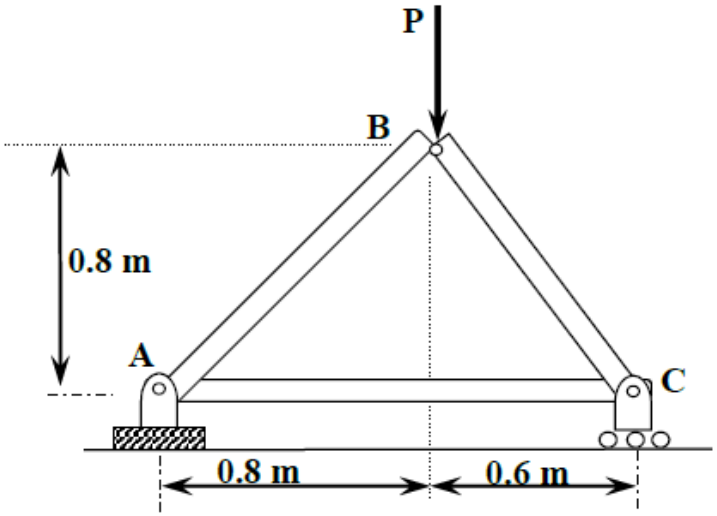


Given l= 3m, E=205GPa

2.7. A small railroad bridge is constructed of steel members, all of which have a cross-sectional area of 3250 mm^2. A train stops on the bridge, and the loads applied to the truss on one side of the bridge are as shown in the sketch. Estimate how much the point R moves horizontally because of this loading.



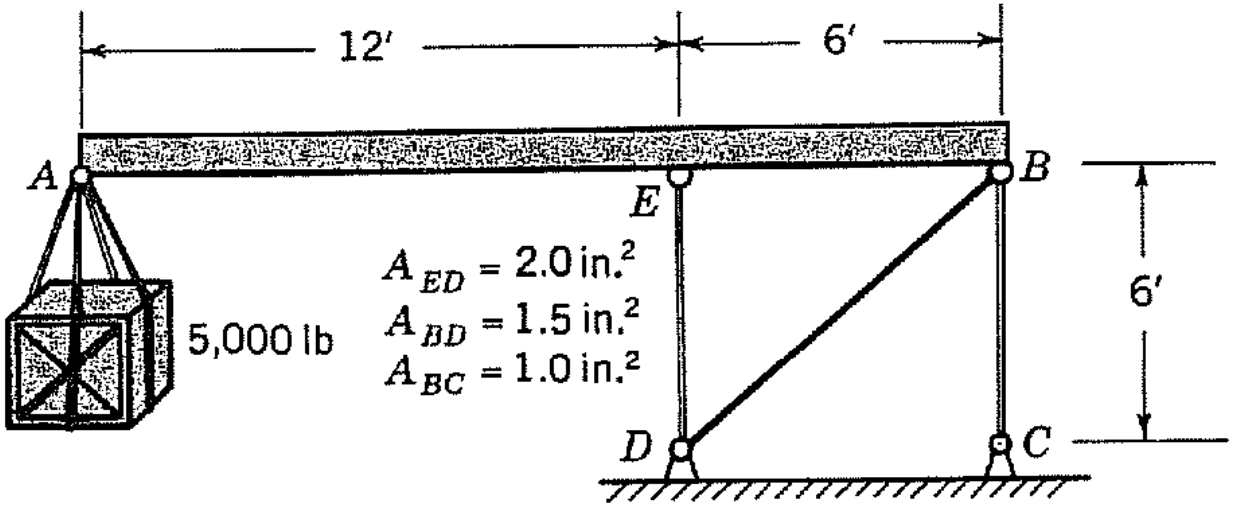
The truss shown in Fig. is made of three steel members, each having a cross-sectional area of 400 mm^2. Applying the energy method (i.e.Castigliano method), determine the magnitude of P required to displace the roller (i.e. Point C) to the right by 0.2 mm. Take E = 200 GPa.



Problem 2.6

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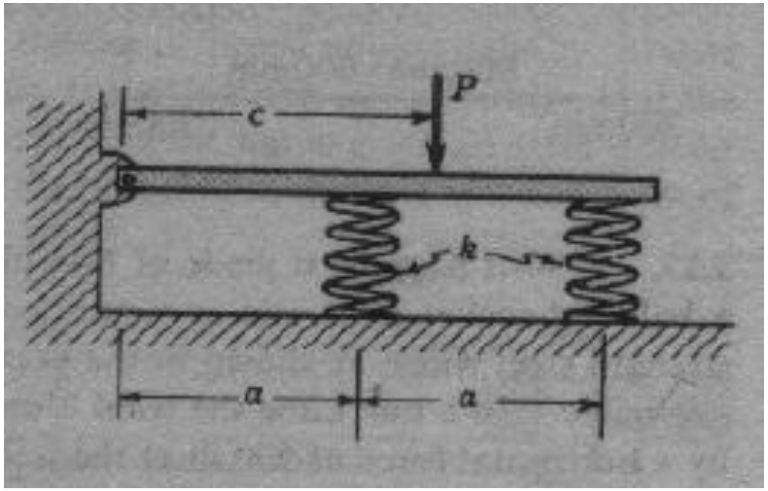
2.6. The stiff member AB is horizontal before the load of 5,000 lb is applied at A. The three steel bars ED, BD, and BC are fastened with pins at their ends. Find (*a*) the force in the bar BD, and (*b*) the horizontal and vertical movement of the point A.



Q2.19

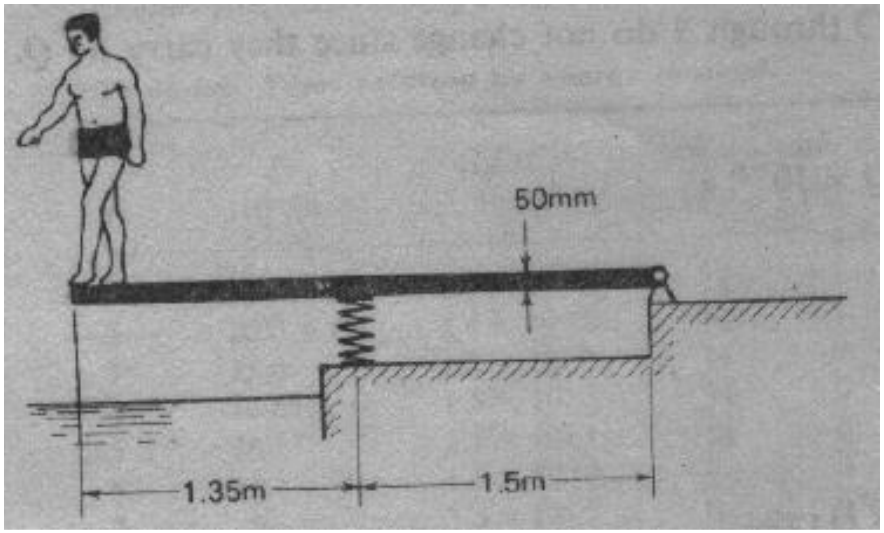
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A stiff beam is hinged at one end and supported by 2 springs of spring constant ‘k’. Where should a force P be applied so that the spring constant of the system (P/δp) is (20/9)k

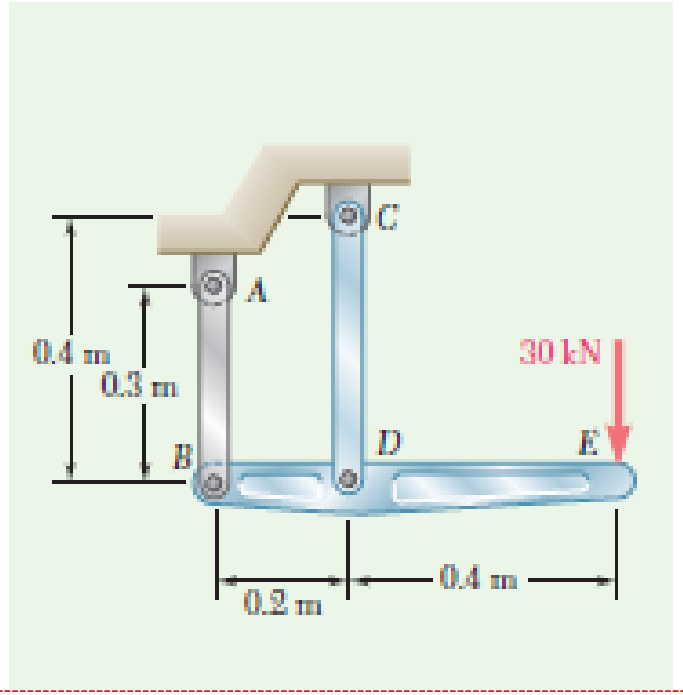


Q2.1

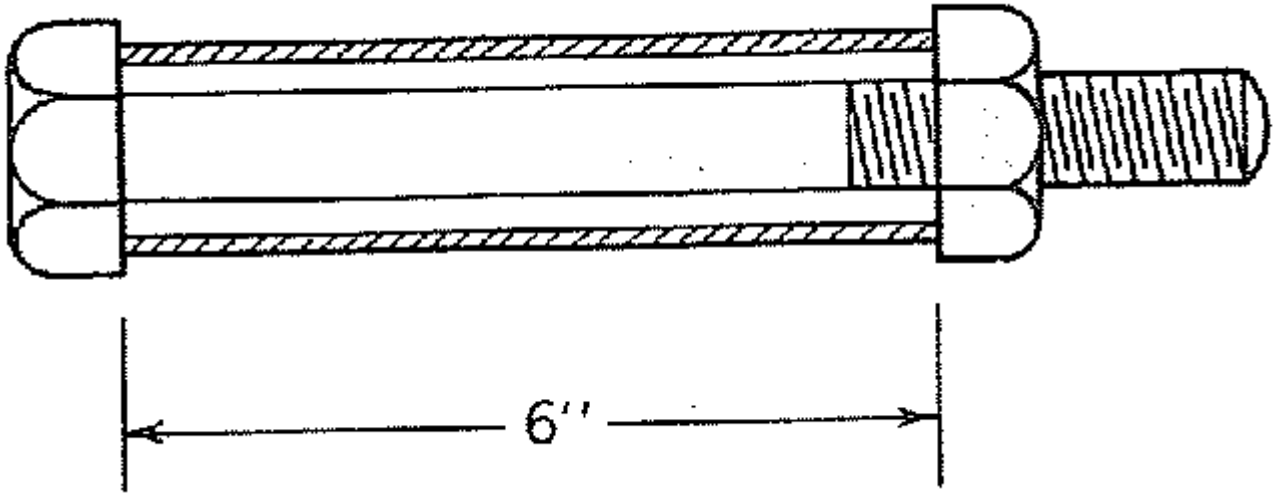
A wood diving board is hinged at one end and supported 1.5m from this end by a spring with a constant of 35kN/m. How much will the spring deflect if a young man weighing 600N stands at the end of the board? Will the deflection be altered if the board is made very rigid.



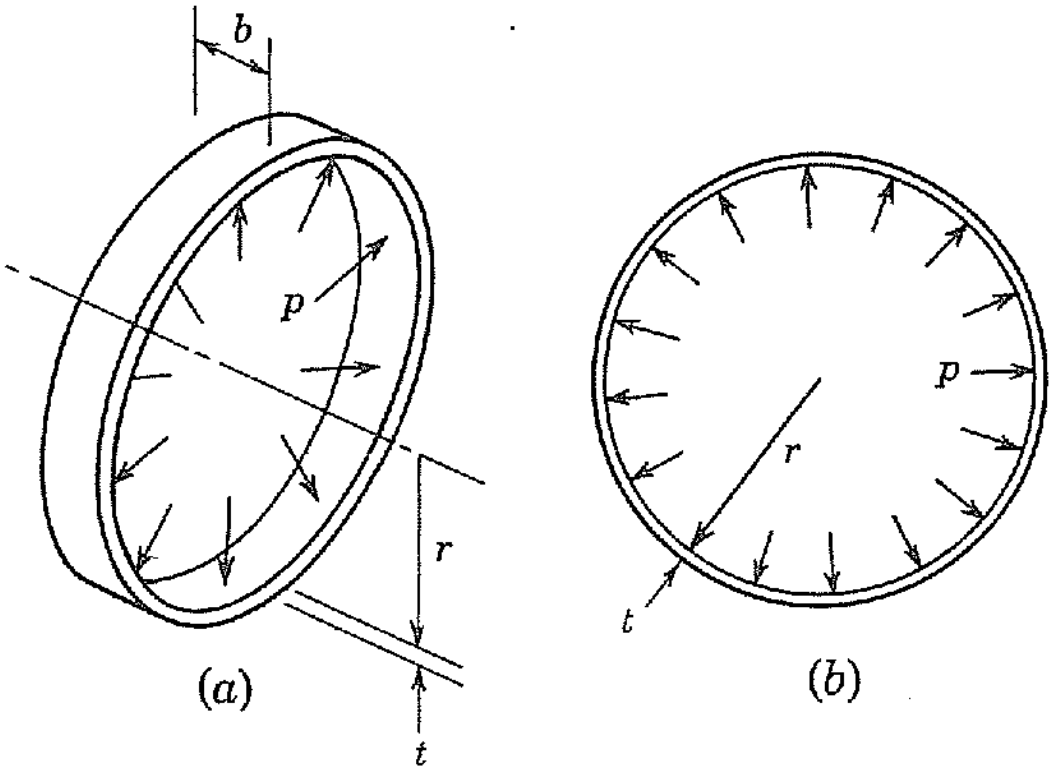
The rigid bar BDE is supported by two links AB and CD. Link AB is made of aluminum ($E = 70$ GPa) and has a cross-sectional area of 500 mm^2. Link CD is made of steel ($E = 200$ GPa) and has a cross-sectional area of 600 mm^2. For the 30-kN force shown, determine the deflection (a) of B, (b) of D, and (c) of E.



Problem 2.24

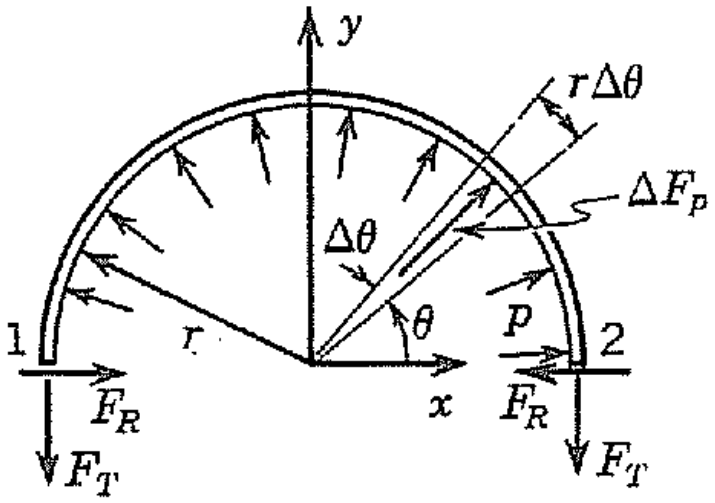


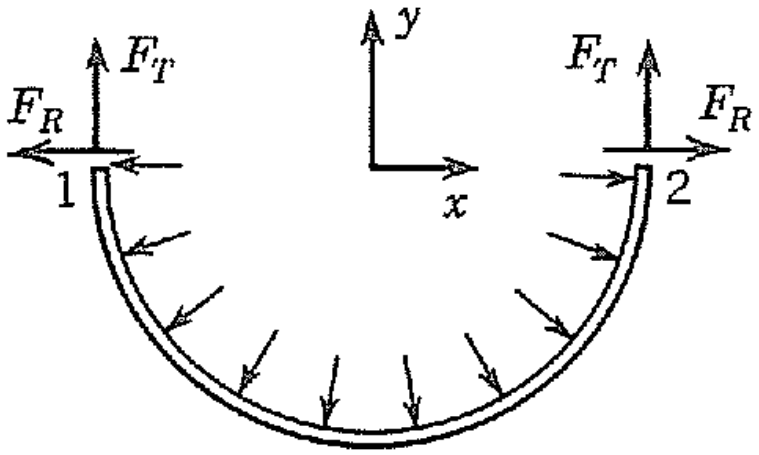
A bolt is threaded through a tubular sleeve, and the nut is turned up just tight by hand. Using wrenches, the nut is then turned further, the bolt being put in tension and the sleeve in compression. If the bolt has 16 threads per inch, and the nut is given an extra quarter turn (90^0) by the wrenches, estimate the tensile force in the bolt if both the bolt and sleeve are of steel and the cross-sectional areas are
Bolt area: 1 $inch^2$ sleeve area: 0.6 $inch^2$, $E=30X10^6$ lb/in^2

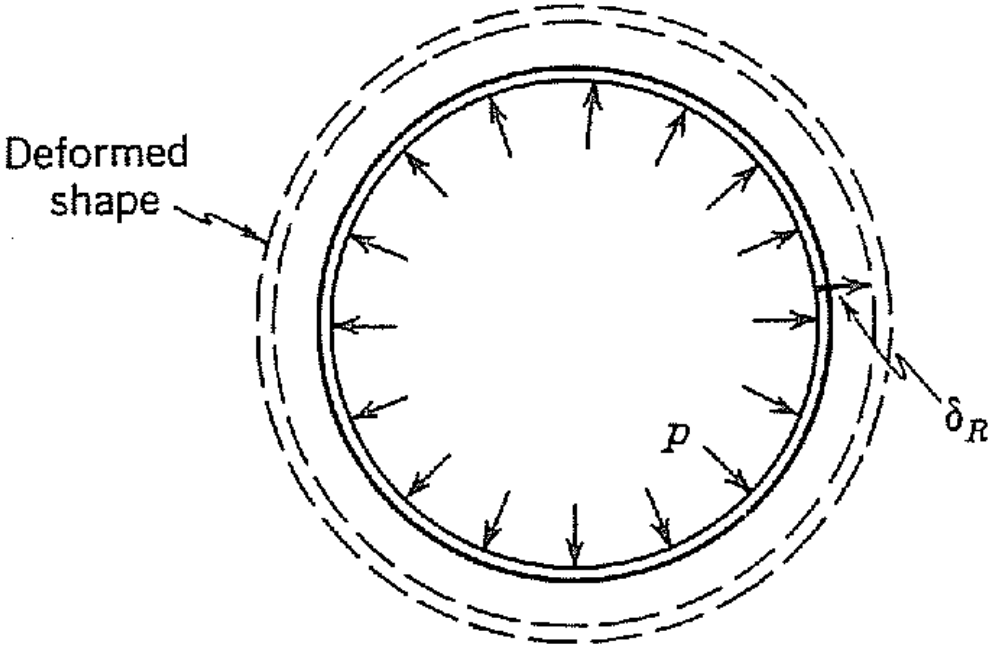


Determine the forces in the ring.

Determine the deformation of the ring due to the internal pressure.


y
$r\Delta\theta$
ΔF_p
$\Delta\theta$
θ
p
1
2
r
x
F_R
F_T
F_R
F_T


y
F_T
F_T
F_R
F_R
1
2
x


Deformed
shape
p
δ_R

References:

Introduction to Mechanics of solids by Crandall
Mechanics of materials by Gere
Mechanics of materials by Beer and Johnston


THANK YOU FOR YOUR ATTENTION